PORTALDOT HACKER HOUSE 2026 · MODULAR CLOUD & MIDDLEWARE

PortalKit

Ship Solidity dApps on Portaldot V3 in minutes.

Scaffolding, diagnostics, a browser dashboard and EVM ↔ Substrate account tools, proven end to end by **PortalVault**, a tokenized-gold vault live on the V3 testnet.

Team **PortalX** · portalkit.cristhedev.com

THE PROBLEM

V3 brings an EVM. Solidity developers still can't find the way in.

No path in

Which RPC, which chainId, where does test POT come from? The top complaints of the last hacker house, and every team rebuilds the same chain config and wallet wiring.

Two account worlds

Tokens arrive at a native `5...` address with 14 decimals; MetaMask shows a `0x` account with 18. New users see an empty wallet and don't know why.

Unfamiliar costs

On pallet-revive a deploy is mostly a code storage deposit. Wallets fail with an opaque "insufficient funds" instead of a number.

Opaque failures

A wrong chainId, a node missing standard `eth_*` methods or an unfunded deployer surface as cryptic tx errors, not a diagnosis.

THE SOLUTION

One toolkit, driven by one config file

`create-portalkit-app`

Foundry or Hardhat + Next.js/wagmi dApp, wired to Portaldot V3, deployable with one command.

`portalkit doctor`

RPC, chainId, the 9 `eth_*` methods tools rely on, runtime, EVM/WASM availability and balances, on both the EVM and Substrate side.

Dashboard

Add network to MetaMask, deploy wizard with a pre-flight tPOT cost estimate, ABI playground with decoded events.

Dual-VM accounts

H160 ↔ SS58 via the real pallet-revive mapping, live native balances, and one-click funding of an EVM account from the Portaldot Extension.

`@portalkit/*` packages

viem/wagmi chain definitions and address strategies generated from `networks.json`. Nothing hardcoded.

PortalVault

Tokenized gold in a KYC-aware ERC-4626 vault with issuer yield, built with the kit as a reference dApp.

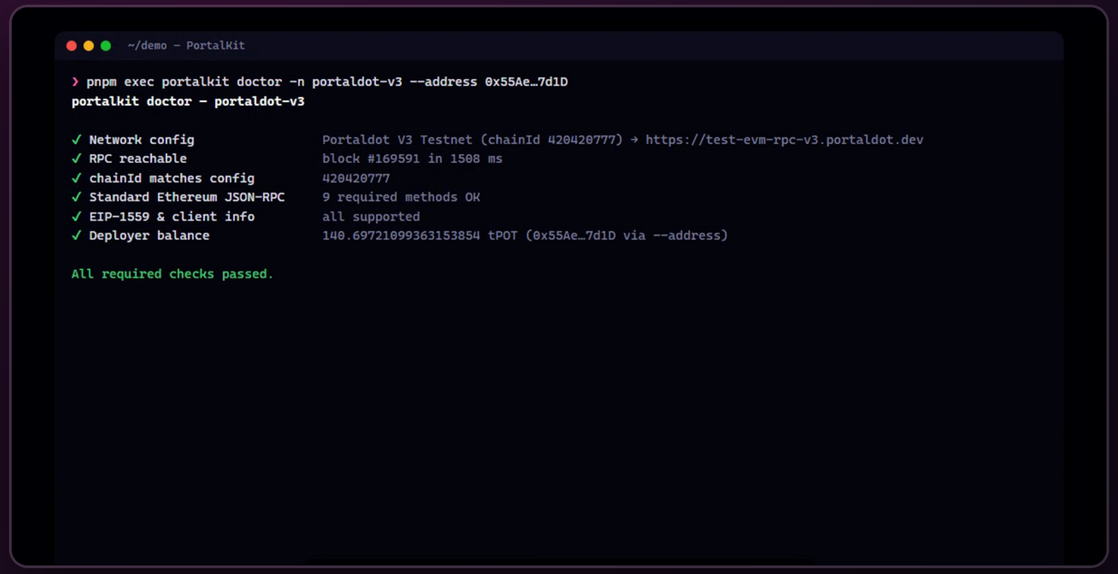
WHY PORTALDOT

Built for what only Portaldot V3 has

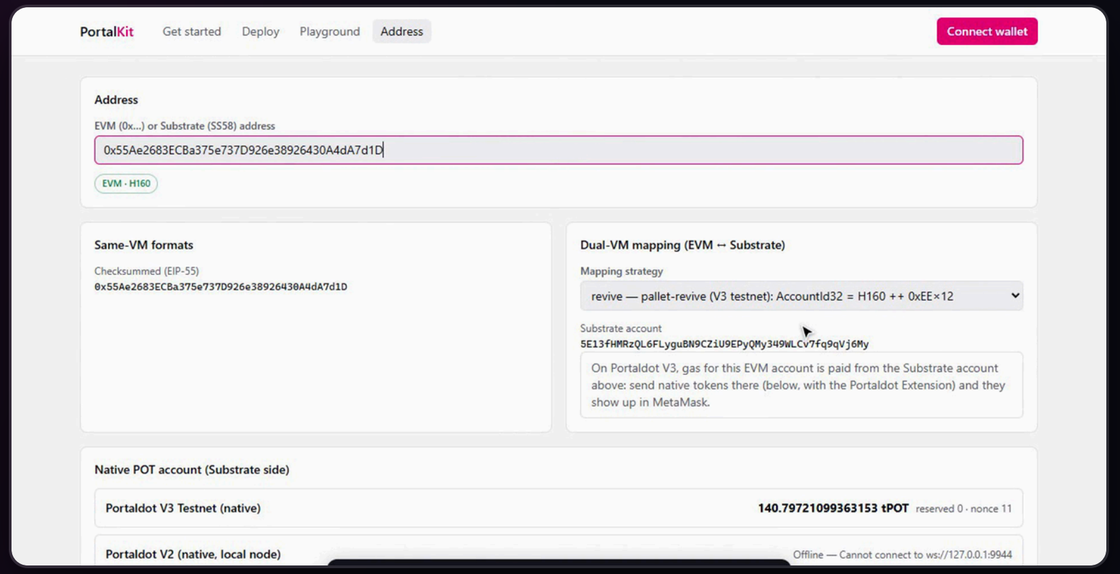
- **Two VMs, one account.** V3 runs EVM (pallet-revive) next to its Substrate runtime. PortalKit is built around that: every EVM address has a native twin that pays its gas, and the kit shows and funds both sides.
- **Doctor understands Portaldot.** It reads `system_properties`, the runtime version and which execution environments a node exposes, and it decodes Portaldot's native account layout.
- **Grows the builder base.** Solidity is the largest smart-contract developer pool. Lowering the cost of their first deploy brings more dApps, more POT usage and more activity to the chain.
- **RWA narrative.** PortalVault shows tokenized assets with compliance (KYC-gated shares, issuer yield) running on Portaldot today.

PROOF · IT WORKS ON V3 TODAY

Real checks, real accounts



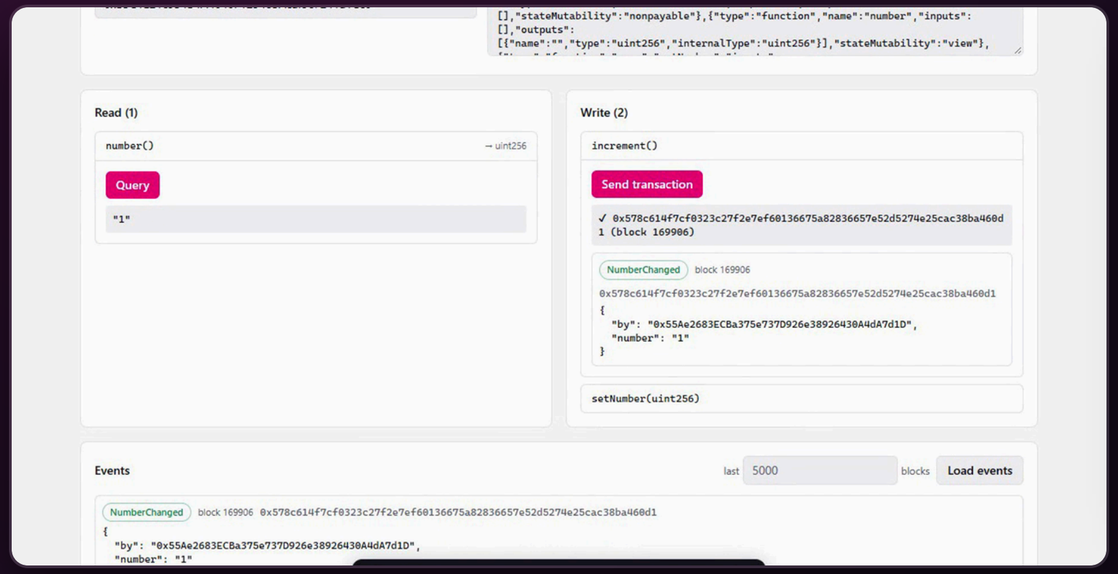
`portalkit doctor` against the live V3 EVM RPC, all checks green



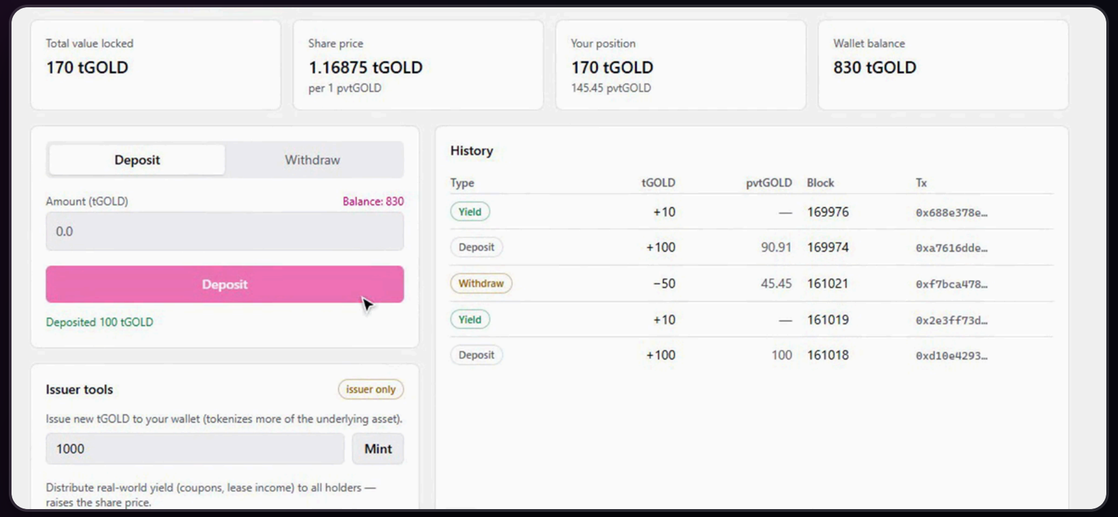
Address converter: EVM deployer → its native gas account and live tPOT balance

PROOF · DEPLOYED ON PORTALDOT V3 TESTNET

Deploy, call, and run a full RWA flow



Contract deployed from the dashboard, `increment()` sent, `NumberChanged` decoded



PortalVault on V3: deposit, issuer yield (share price 1.16875), withdraw, on-chain history

MockRWA (tGOLD)	`0x1ac55f7161efa932aa6963e86fac92b74d45094d`
RWAVault (ERC-4626)	`0xe5378c971cf26453926b941dabf247cf1fd44697`

TRACTION & QUALITY

Not a mock-up

20

real V3 testnet transactions (EVM + native), every hash in `docs/evidence.md`

2

live apps: dashboard and PortalVault, both reading Portaldot V3

150+

automated tests: 60 Foundry (fuzzing, inflation attack) and 91 Vitest, plus browser E2E

- Both scaffold presets (Foundry, Hardhat 3) verified end to end; the Foundry preset deployed a contract to V3.
- Config-driven: when V3 changes an RPC or chainId, one JSON file updates every tool.

WHY KEEP SUPPORTING IT

Infrastructure every Portaldot EVM builder needs

Next 4 weeks

Publish `@portalkit/*` to npm so `npx create-portalkit-app` works anywhere · native SS58 → H160 mapping · Hardhat preset verified on V3

With mainnet EVM

Contract verification and explorer links · mainnet presets · hosted dashboard as a public good for the community

Dual-VM

If V3 exposes EVM ↔ native/WASM precompiles: one account view with both balances, and calls across VMs from the playground

Beyond Solidity

ink!/WASM templates next to the Solidity ones, so one kit covers both of Portaldot's execution environments

PortalKit

dashboard	portalkit.cristhedev.com
vault	portalkit.cristhedev.com/vault
code	github.com/cris-the-dev/portaldot-hackathon
video	youtu.be/89P2RE-D5c8

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